2D Array

→ Array of arrays

→ Collection of arrays

datatype arrayname[rows][cols];

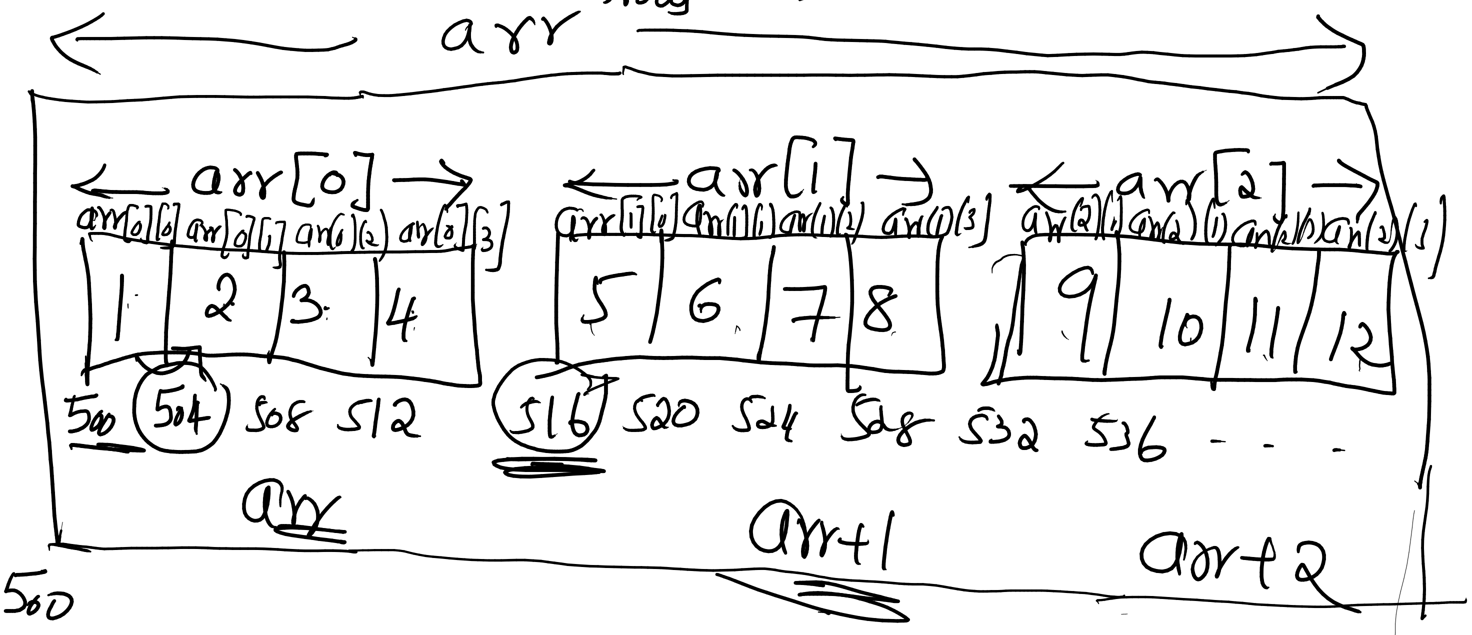
cols → no of elements in each 1D array

rows → no of 1D arrays

rows → col

int arr[3][4]; each 1D ~~has~~ array → 4 elements

3 → rows


arr
arr[0]
arr[1]
arr[2]
arr[0][0] arr[0][1] arr[0][2] arr[0][3]
arr[1][0] arr[1][1] arr[1][2] arr[1][3]
arr[2][0] arr[2][1] arr[2][2] arr[2][3]
1 2 3 4
5 6 7 8
9 10 11 12
500 504 508 512
516 520 524 528 532 536 - - - -
arr
arr+1
arr+2
500

rows => 3 → 0, 1, 2 arr[i]th 1D array

→ i = 0 → j → 0 1 2 3

i = 1 => arr[1] → 0 1 2 3 → arr[1][0] arr[1][1] arr[1][2] arr[1][3]

**a+1 j=1

"%u", arr // 500

"%u", *arr // *(500) // 1D array

"%d", **arr; // 2 times dereferencing => data/element

arr[0][0] = 1

Pf(" /.u",(* arr)+1)

↳ 1 time dereference // 1D array

*

+

arr[0] + 1 x sizeof(int)

500 + 4 => 504

~~*~~**arr)+1

arr[0][0] +1

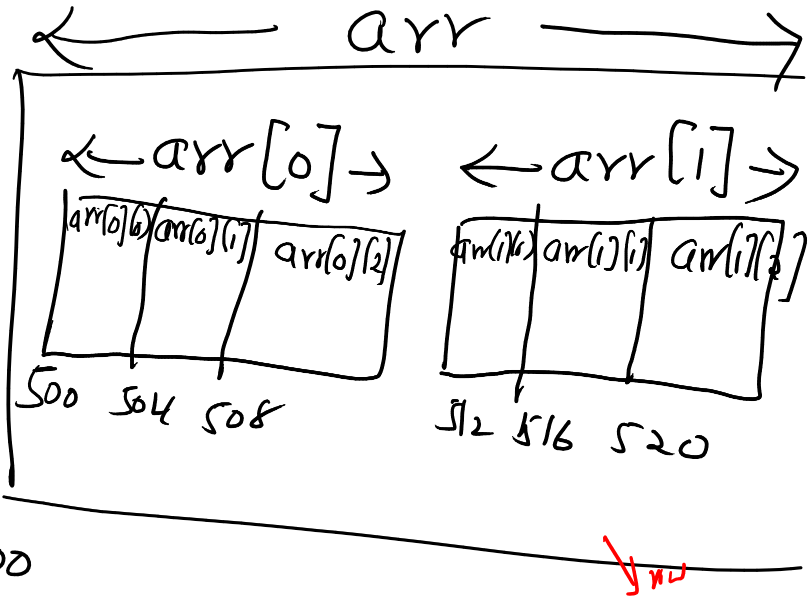
↕

1 +1 => 2

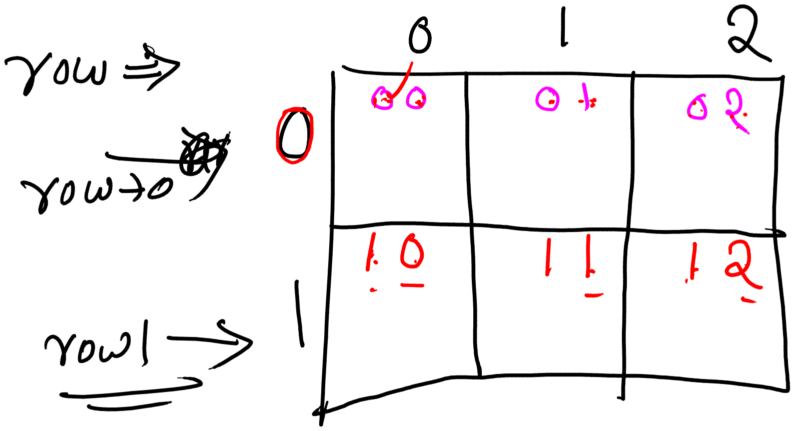
rows

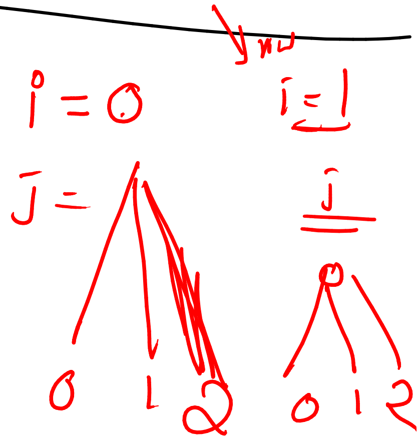
col

int arr[2][3] elements in each 1D array



500

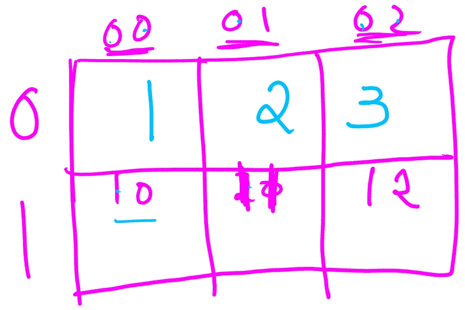




i=0, j= 0, 1, 2, ~~3~~

arr[0][0], arr[0][1], arr[0][2]

```
for(i=0; i<2; i++)
{
    for(j=0; j<3; j++)
    {
        scanf("%d",&arr[i][j]
    }
}
```



```
for(int i=0; i<2; i++)
{
    for(j=0; j<3; j++)
    {
        Scanf("%d",&arr[i][j
    }
}
```

Varying one make it as inner loop

i=0, 0<2 → outer loop

j=0; j<3 (0<3 ✓

Scanf ("%d",&arr[i][j]);

}

i → arr[0][0] → j=0

arr[0][1] → j=1

arr[0][2] → j=2

j=3 X

i=1, 1<2

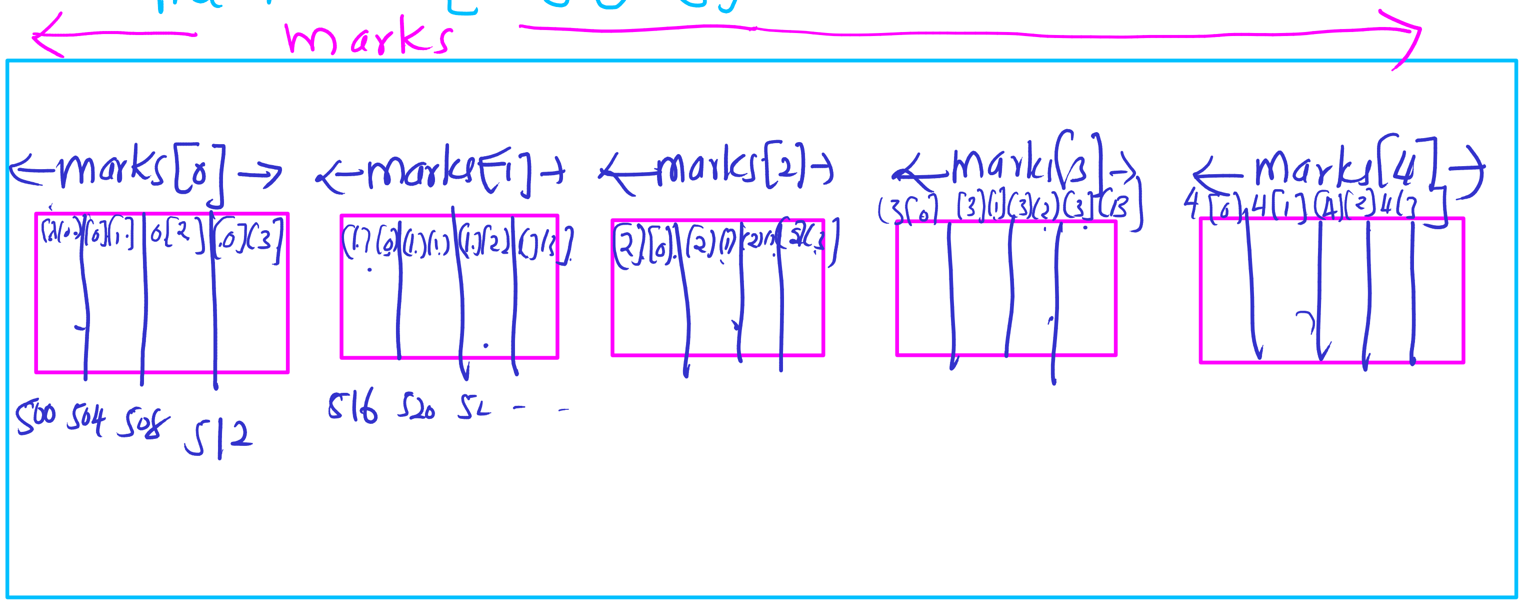
Scanf("%d",arr[i][j] → arr[1][0]

arr[1][1]

arr[1][2]

---

int marks[5][4];



500

Student

r.w →

	0	1	2	3	Total marks
0	00 10	01 20	02 80	03 90	100
1	10 45	11 65	12 85	13 35	25
2	20 10	21 10	22 20	23 30	85
3	30 45	31 85	32 95	33 100	45
4	40 10	41 20	42 85	43 65	70

⇒ marks [5] [4]

i = 0
1
2
3
4
~~5~~

outer loop ⇒ constant
↳ 5

i = fixed

j : i vary

```
for(int i=0; i<5 ; i++)
{
    for(int j=0;j<4;j++) → j=0
    {                          1
                               2
                               3
        Scanf("%d", &marks[i][j]); 4×
    }
}
```

mark [0][0]
[0] [1]
[0] [2]
[0] [3]

marks[0][0]+[0][1]+[0][2]+[0][3]

marks [5] [5]

row 0 col 0, row 0 col 1, row 0 col 2, row 0 col 3, row 0 col 4

Row

row 0 → 00 + 01 + 02 + 03 ⇒ 04

row 1 → 10 + 11 + 12 + 13 ⇒ 14

row 2 → 20 + 21 + 22 + 23 ⇒ 24

row 2 → 30 + 31 + 32 + 33 → 34

row 4 → 40 + 41 + 42 + 43 → 44

```
for(i=0;i<=4;i++)    i<5 or
{
    for(j=0;j<4;j++)
    {
        Sum=Sum+Mark[i][j];
    }
    ~~Sum=~~ arr(i)(j)=Sum;
```

i=0 , Sum=0

j

0 1 2 3 (j<=3) or j<4

	j= 0	1	2	3	4
1st → i= 0	0 0 1	0 1 2 ✓	0 2 3	0 3 4	0 4 10
2nd → 1	10 2	11 2	12 2	13 2	14 8
2	20 3	21 3	22 3	23 3	24 12
3	30 4	31 4	32 4	33 4	34 16
4	40 5	41 5	42 5	43 5	44 20

~~0 1 3~~ ~~6~~ 10

Sum

i=0; i<5 ✓

j=0, 1, 2, 3

(j=0; j<4; j++)

0<4✓

Sum=Sum+mar[i][j]

0+mark[0][0]

1

j=1, 1<4

Sum=Sum+m[i][j]

=1+m[0][1]

sum=1+2

j=2, 2<4

Sum=sum+m[i][j]

sum= 3+m[0][2]

sum=3+3

j=3, 3<4

Sum=Sum+m[i][j]

=Sum+m[0][3]

Sum=6+4

j=4, 4<4 X

i=0, j=4

```
int i, j;
for(i=0; i<5 ; i++)
{
    Sum=0;
    for(j=0; j<4; j++)
    {
        Sum=Sum+mar[i][j]
    }
    ~~marks~~s[i][j] = Sum;
}
```

```
topmark = mark[0][4]; →
            ind = 0;        i=0 X

for (i=1; i<5; i++)
{
    if (mark[i][4] > topmark)
    {
        topmark = mark[i][4]
        ind = i;
    }
}
```

[0][4]

[1][4]

[2][4]

[3][4]

[4][4]

topmark = toppech

ind = student